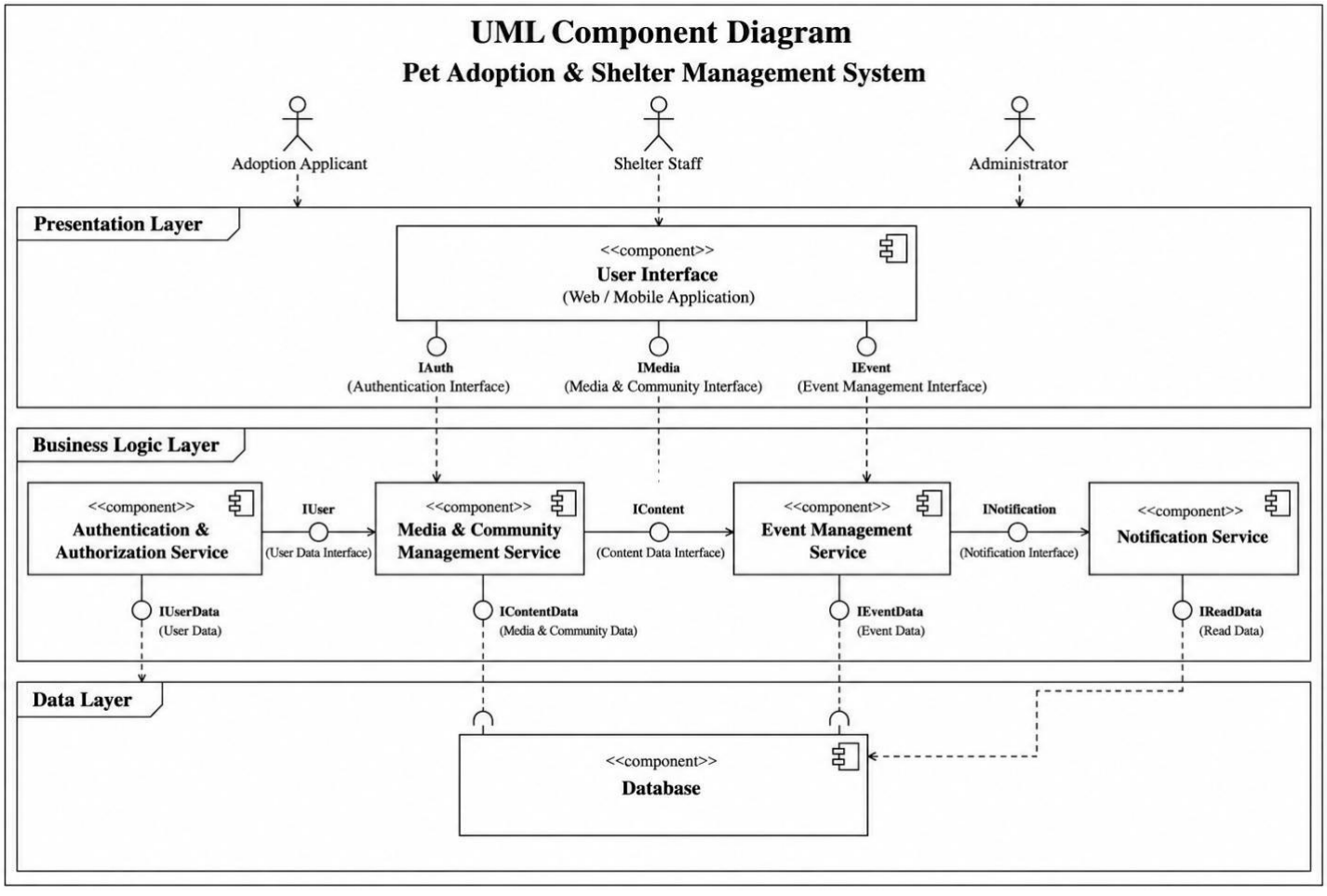
NAME : DEERAJ S DATE : 08/09/26

SRN : PES2UG25CS806 LAB : 03

TOPIC : Pet Adoption & Shelter Management System

DELIVERABLE - 01

1. Component Diagram (PNG or PDF)
   Your diagram must include atleast 5 components (Adding attributes is optional). Interface Requirements:
   - Show atleast 4 interfaces between components:
   - Use proper UML notation for provided/required interfaces
   - Show data flow and component interactions



DELIVERABLE – 02

Written Justification

Architecture Selection

We chose **Layered Architecture** for the **Pet Adoption & Shelter Management System**. The system is divided into separate layers for presentation, business logic, and data management. This provides a clear structure for handling user interactions, system operations, and data storage.

Architectural Choice – Reasons

1. Clear Separation of Responsibilities:
Layered Architecture separates the user interface from the core business functions and database operations. In our system, components such as **User Interface, Authentication & Authorization, Media & Community Management, Event Management, Notification, and Database** can perform their specific responsibilities independently. This makes the system easier to understand, develop, test, and maintain.

2. Easier Maintenance and Modification:
The architecture allows individual components to be modified without requiring major changes to the entire system. For example, changes to the **Event Management** functionality, such as event registration and checking available registration slots, can be handled within the business layer without directly affecting the presentation or database layers. This supports the requirements defined for event management and registration.

Security Advantage

Layered Architecture allows **authentication and authorization** to be handled as a separate component in the business layer. This helps control access to sensitive operations and ensures that only authorized **Shelter Staff and Administrators** can upload, publish, update, or moderate content. This directly supports the system's security requirement that unauthorized Adoption Applicants must be denied access to content-management operations.

Performance Benefit

Separating the business logic from data access allows the system to efficiently retrieve and process **Media, Events, and Community** information. This can help provide faster responses to users and supports the requirement that the Media, Events & Community pages should be displayed within **3 seconds under normal operating conditions**.

Conclusion

Overall, Layered Architecture is appropriate for our system because it provides **clear organization, easier maintenance, controlled access, and efficient data handling** while supporting the functional and non-functional requirements of the Pet Adoption & Shelter Management System.